SOC Lab Report: Defense Evasion Detection using Splunk

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1 Abstract

This report demonstrates detection of defense evasion techniques in a SOC environment using Splunk. Attackers often attempt to remove evidence of their activity by clearing logs. This scenario simulates log clearing and detects it using Windows event logs.

2 Objective

To detect log clearing activity using Splunk and analyze defense evasion techniques.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Defense Evasion
- Technique: Indicator Removal (T1070)

5 Attack Simulation

Log clearing was simulated using Windows command-line utility.

5.1 Command Used

```
wevtutil cl Security
```

5.2 Explanation

The command clears all security event logs, removing evidence of previous activity.

5.3 Screenshot

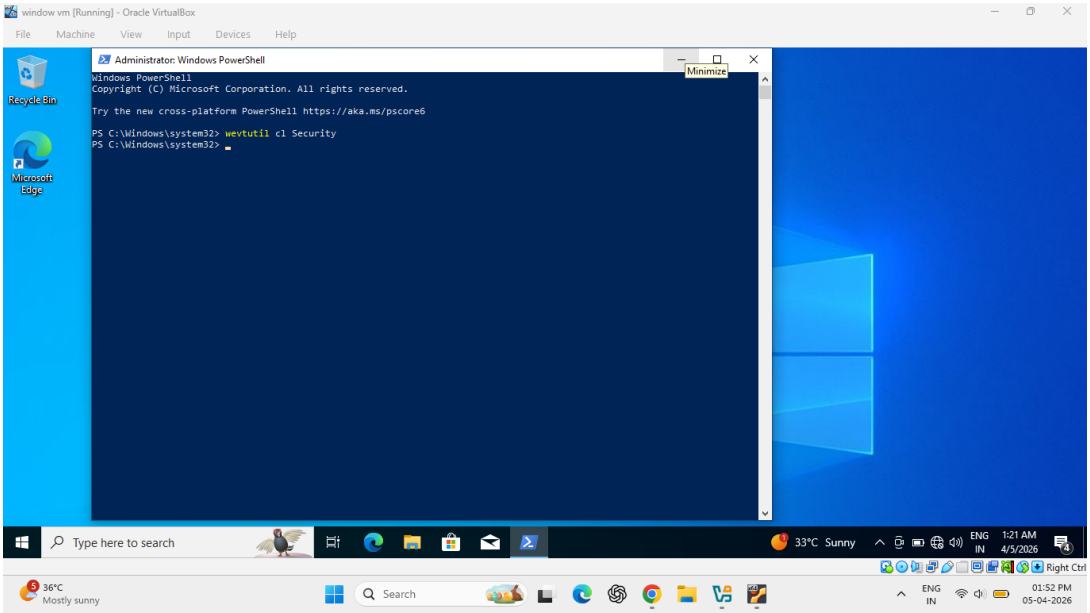

Figure 1: Clearing Windows Security Logs

6 Detection in Splunk

Log clearing is detected using EventCode 1102.

6.1 Detection Query

```
index=windows EventCode=1102
| table _time, user, host
```

6.2 Explanation

EventCode 1102 indicates that the security log has been cleared.

6.3 Screenshot

7 Investigation

Further investigation was performed to identify the source of log clearing.

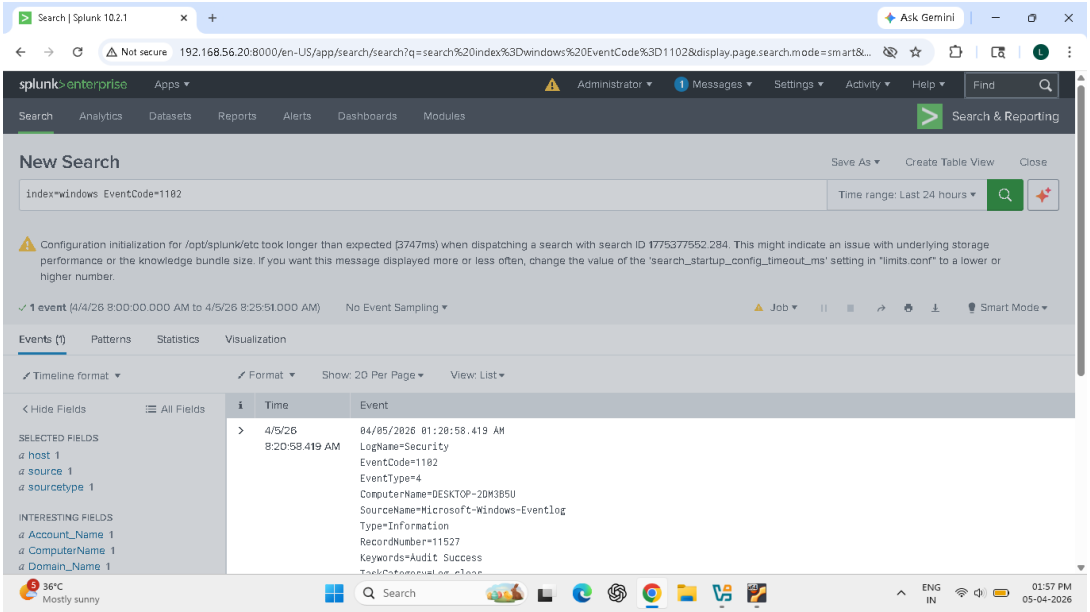

Figure 2: Detection of Log Clearing Activity

7.1 Investigation Query

```
index=windows EventCode=1102
| stats count by user, host
```

7.2 Timeline Analysis

```
index=windows user="TARGET_USER"
| sort _time
```

7.3 Findings

- Security logs were cleared
- Activity performed by a specific user
- Possible attempt to hide malicious actions

7.4 Screenshot

8 Response Actions

Immediate actions were taken due to high severity.

- Isolated the affected system
- Disabled the user account
- Escalated the incident

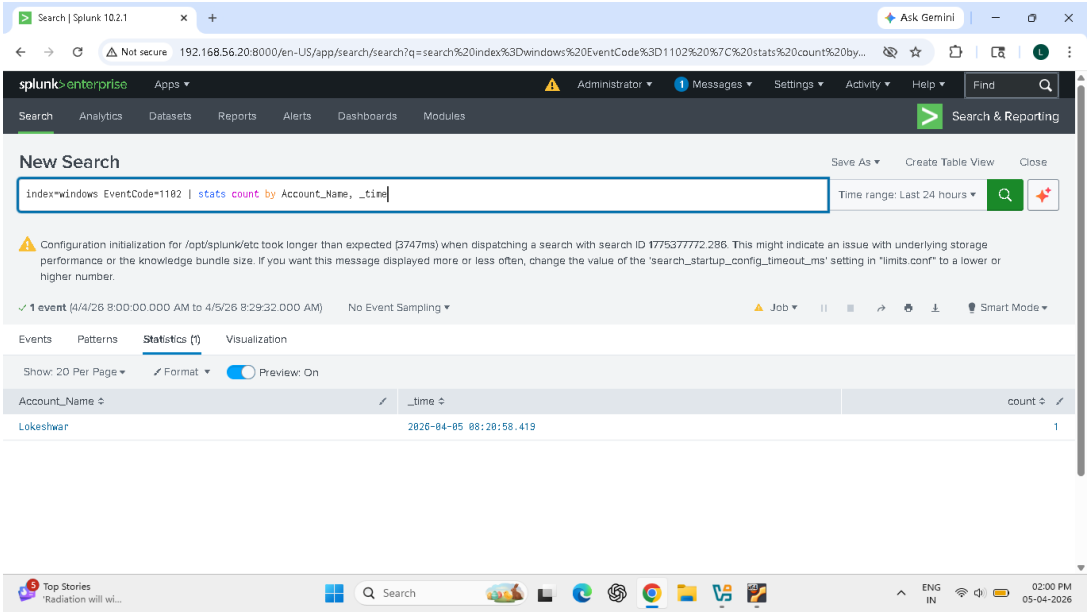

Figure 3: Investigation showing log clearing activity

8.1 Commands Used

```
netsh interface set interface "Ethernet" admin=disable
net user <username> /active:no
```

8.2 Explanation

The system was isolated to prevent further activity, and the user account was disabled to stop unauthorized actions.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

Security logs were cleared, indicating a defense evasion attempt to remove evidence of malicious activity.

9.2 Impact

Loss of critical forensic data and reduced visibility into attacker actions.

9.3 Conclusion

The activity was successfully detected and mitigated using Splunk SIEM.

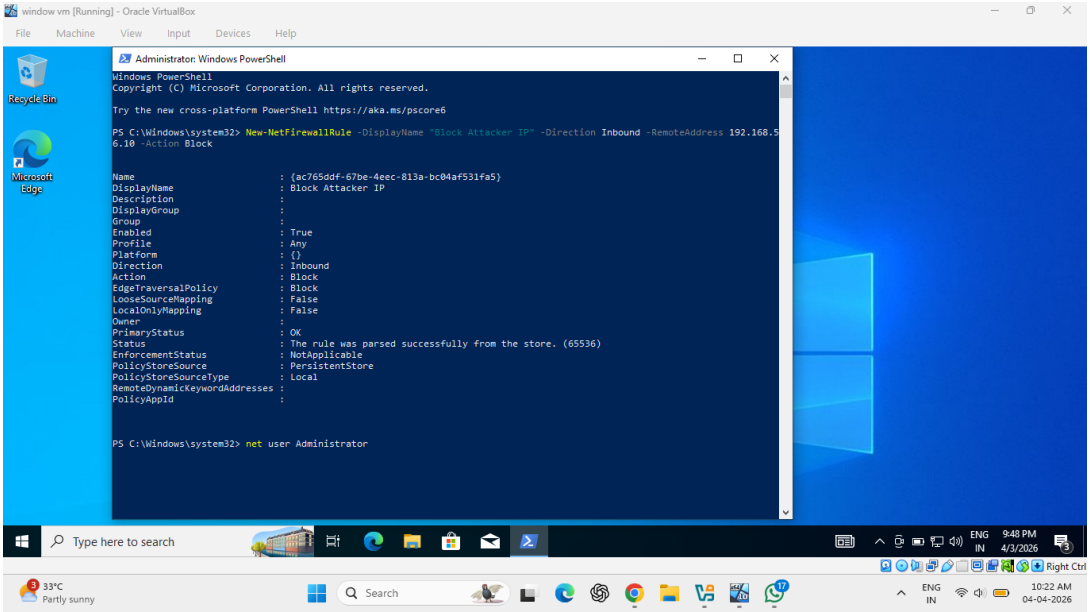

Figure 4: Response Actions - System Isolation and Account Disable

10 Conclusion

This scenario highlights the importance of monitoring log integrity and detecting attempts to remove evidence.